

Homework 8

ASTR 5900, Computational Physics & Astrophysics

Due May 8, 2026, 11:59 PM

Objective

Train a denoising autoencoder to recover clean sine-wave signals from noisy inputs, and investigate how noise type, training size, and network architecture affect reconstruction performance.

Required: `pip install tensorflow`

Reference notebook: [hw8.reference.ipynb](#) - A starter notebook covering Keras syntax and model training. You may build on it or start your notebook from scratch.

Denoising Autoencoder for Sine Curves

Q1. Data Generation

(2 points)

Generate 10,000 sine curves $x(t) = A \sin(\omega t + \phi)$ with

$$A \sim \mathcal{U}[0.5, 1.5], \quad \omega \sim \mathcal{U}[0.5, 2], \quad \phi \sim \mathcal{U}[0, 2\pi]$$

sampled at $N = 100$ equally spaced points over $t \in [0, 4\pi]$. Add Gaussian noise:

$$x_{\text{noisy}}(t) = x(t) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \quad \sigma = 0.2$$

- Plot four clean/noisy curve pairs.
- Split into 90% training / 10% validation (input: noisy, target: clean).

Q2. Autoencoder Model

(3 points)

Train a fully connected autoencoder with architecture

$$N \longrightarrow 64 \longrightarrow 16 \longrightarrow 64 \longrightarrow N, \quad N = 100$$

using mean squared error (MSE) loss.

- Plot training and validation loss vs. epoch.
- Plot clean / noisy / reconstructed signals for at least four validation examples.

Q3. Parameter Exploration

Investigate how performance changes when modifying the following parameters. For each experiment, provide plots of training and validation loss vs. epochs. Also report the architecture and comment on whether training converges and whether you observe overfitting.

(a) **Training data size** (1 point)

- 90% training , 10% validation (baseline)
- 60% training , 40% validation
- 30% training , 70% validation

Discuss the effect on validation MSE and overfitting.

(b) **Network architecture** (1.5 points)

Fix the bottleneck at 4 and experiment with different numbers of layers and neurons.
For example:

Model 1 (Baseline): $100 \rightarrow 64 \rightarrow 4 \rightarrow 64 \rightarrow 100$

Model 2 : $100 \rightarrow 80 \rightarrow 40 \rightarrow 4 \rightarrow 40 \rightarrow 80 \rightarrow 100$

Model 3 : $100 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 4 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 100$

Record the training time and final validation MSE for each architecture.

(c) **Bottleneck size** (1.5 points)

Fix the outer architecture ($N \rightarrow 64 \rightarrow \cdot \rightarrow 64 \rightarrow N$) and vary the bottleneck dimension: 16, 8, 4, 3, 2, 1. Determine the smallest bottleneck size that still reconstructs the signal accurately.

(d) **Noise models** (1 point)

Instead of Gaussian noise, test **Laplace noise**.